

Detecting Liver Cancer from Online Search Logs

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Abstract— There are many incurable diseases around us like various types of cancers which are detected at the last stage. But they are curable if early detection is possible. Web search log with symptoms can be effective for such early detection. Online search queries can act as a useful evidence of user's health condition as they search with symptoms in search engine. In this paper, we have used machine learning techniques to identify those attributes from online search log that can be used for early detection of liver cancer. Liver cancer is diagnosed very late most of the time. Primary stage symptoms are obscure and vague in most of the cases. It becomes very difficult to get the treatment properly because of the metastasis of the cancer when diagnosed. We have proposed a method to organize online search log data to detect liver cancer early. If we have online search log data containing symptoms searched on search engine and the time of the searches of each symptom, they can be easily organized by our method of dataset creation and detect liver cancer. We have estimated the performance of Support Vector Machine (SVM) based on different kernel scale values. The best classification accuracy for $\gamma = 0.03$ is 94.30%. We have also compared the performance of different machine learning techniques. Among them, Random Forest has the highest classification accuracy of 97.50%.

Keywords— *Liver Cancer Detection, Online Search Log, Classification, Support Vector Machine, Naïve Bayes, Neural Networks, Random Forest, Decision tree, k-Nearest-Neighbors*

I. INTRODUCTION

Health conscious people search online with their health-related symptoms first to detect the health problems [1]. Previous researches show that how people search online about their health issues and what kind of query people generally use to find the solution of their health problems [2]. People generally search with their symptoms on search engine to understand the probable underlying causes and the diagnosis of their diseases [3], what are the treatment and medications, the side effects of medications [4], what kind of physician they should consult with. Former studies show how data of social media can be used to detect health problems [5].

Cancer is a leading universal health problem [6]. In both under-developed and well-developed countries, it is the prominent cause of death. [7]. It is in the 2nd position for human death in US. [6]. With the increase in population, the concern will rise all over the world. Specially the concern is more in less developed countries as about 82% of world's total population dwell in there. There are many lifestyle issues known as risk factors poor diet, less physical activity, smoking, reproductive changes. They are severe in less developed countries [7].

In 2000, among the most common cancer, Liver cancer was in 5th position. It was account for about 551,000 new cases. Total number of men was around 399,000 and women was 153,000. 83% of the cases were from developing countries [8].

Only china was responsible for 54% of the entire cases [9]. During the years between 1988 and 1992, the standardized incidence rates of the world for men was 72.1 and women was 19.1 per 100,000 in Quidong [10].

In 2016, the amount of cancer cases was 1,685,210 and 595,690 of them died in US [6]. By 2030, liver cancer deaths will increase so rapidly so that it is expected to be the 3rd leading reason of deaths because of cancer [11].

Cancer which begins in the liver is known as liver cancer. It can also develop and spread to the outside the liver. The location of the liver is below the right lung and under the ribcage. It is one of the hugest organs of our body. It performs several functions.

It removes toxins from our body. There can be malignant hepatic tumors in or on the liver if there is liver cancer. The primary stage symptoms are nonspecific most of the time. That's why it becomes very tough to detect this disease at the right time. In most of the cases it becomes very late to diagnose. For detecting early, online search log can be effective.

In this paper, we have shown that if we have an online search log, we can easily organize them using our dataset creation process and detect whether a user can be a liver cancer patient or not. We have used different machine learning techniques for this purpose and estimated the performance of SVM for different kernel scale values and then compared the performance of all techniques on our dataset.

II. RELATED WORK

Internet is vital to find information about health issues for several people. Previous researches have been performed on how people get health information online. There are various types of procedures to inspect online health search and analyses of large-scale web search queries is one of them [12]. A study of 2013 shows that 59% of American adults searched their health-related problems online. 35% of them were involved in self-diagnosis. More than 50% of these individuals consulted with physicians after taking decisions based on the online search [1]. White and Horvitz discovered that using online search for self-diagnosis, anxiety of about 40% of the online health-problem-searchers has increased because of biased information [13]. It hampers the study of how people use web for health problems, patterns of querying and content having solutions provided in response. A lot of researches have been performed for analyzing search behavior from online search log. There are many applications of search behavior analysis like result ranking [14] and to make prediction future possible searches and interests [15].

III. DATASET

We worked with a balanced dataset based on liver cancer. There are 60% data which are positive that means they are actual patients and 40% are negative that means not patients. In our dataset, there are 400 user ids as samples and 34 attributes containing symptoms of liver cancer and the time of their searches by the users. There are 16 symptoms taken as attributes and all of them have a time when they were searched. The symptom attributes contain binary values so that if an id searches a symptom, the value of the symptom attribute will be 1 and there are attributes called 'days' with each symptom. For an id, the value of 'days' attribute for first searched symptom is taken as 1 and the value of 'days' attributes of other symptoms are taken as the differences in days from the first searched symptom. The value of the 'days' attributes are integer (total number of days differ from first symptom). We have taken 4 stages (1st, 2nd, 3rd and the last stage). Each stage contains 4 symptoms. There is another attribute called 'order' such as if an id searches the symptoms in a sorted order according to the stages of the liver cancer, there will be a value 1 for this kind of searches. Otherwise, there will be a 0 if user searches in a random time manner. Here, order means the symptoms of 1st stage are searched before 2nd stage, 2nd stage before 3rd and 3rd stage symptoms are searched before last stage. At last, we have an attribute called 'Class' which contains two types of values 'yes' and 'no'. Containing 'yes' means actual patient and we have split the 'no' value in 3 types random searcher, medical student and Other disease's symptoms searcher.

A. Actual Patients

If an id is an actual patient, the attribute will be yes otherwise no. An id can be a patient who searches in a sorted manner according to stages of the liver cancer. So, value of the order attribute will be 1. Total 60% of the ids in the dataset are actual patients.

B. Random Searchers

They can be random searcher who searches in different times but not in a sorted way. They can search last stage symptoms at first. So, value of the order attribute will be 0. Total 20% of the ids in the dataset are random searchers.

C. Medical Students

Ids can be medical students with having the class attribute as no if the search time of the ids are limited to some months as their semester progresses. Medical students can search both in order or not in order. So, value of the order attribute will be 1 or 0. Total 15% of the ids in the dataset are medical students.

D. Other disease's symptoms searchers

There can be another type of ids which will have 'no' in class attribute. They are the ids who have other diseases but search the symptoms common with liver cancer. They will search in an order but will not be a liver cancer patient. So, value of the order attribute will be 1. Total 5% of the ids in the dataset are Other disease's symptoms searchers.

IV. METHODOLOGY

A. Support Vector Machine(SVM)

The Support Vector Machine a well-developed technique to solve binary classification problems [16]. Multi-class problems($k > 2$ classes), are generally figured out by decomposing them into several binary class problems [17]. It takes $N(N-1)/2$ binary SVMs for an N -class problem if we use 'one-against-one' procedure. On the other hand, 'one-against-the others' procedure needs N SVMs [18]. 'One-against-one' procedure shows better performance but to solve many real-world problems it takes excessively expensive computing resources [18]. We have worked with a binary classification problem and for that we have used 'One-against-one' approach in this paper. For binary classification($k=2$), single constraint is used for each training point.

$$(w_{y_i} - w_{1-y_i}) \cdot x_i + b_{y_i} - b_{1-y_i} \geq 1 - \varepsilon_i^{1-y_i}$$

Here, we label $y \in \{0,1\}$ [19]

A group of mathematical functions are used in SVM. They are known as kernel. SVM executes two- dimensional classification with the help of kernel function. We have used Radial Basis Function (RBF) in this work. The equation for RBF is:

$$K(x, x_i) = e^{-\gamma \|x - x_i\|^2},$$

Kernel scale $\gamma > 0$ [20]

B. k-Nearest-Neighbor(kNN)

The kNN is considered as non-parametric procedure [21]. It searches a bunch of k samples and those samples are closest to unrevealed samples (based on Euclidian distance in our work). Unrevealed sample's labels (class) are decided by computing the average of the class attributes from these k samples [22]. Value of k has a great importance in estimating the performance of the classifier and main tuning parameter is k [23]. We have used $k=5$ in this paper.

C. Random Forest

While implementing Random Forest [24], we need to set up two parameters. One is the quantity of trees and another one is the total of features in each split. Previous researches show that using default parameters we can get acceptable results [25]. However, Liaw & Wiener stated [25], large number of trees gives better results. But, According to Breiman [24], though excess number of trees than required does not harm the model, they can be irrelevant.

D. Naïve Bayes Classifier

It is a simple method having probabilistic knowledge along with clear semantics. It is formulated to solve supervised induction problems. The execution is quantified based on the accuracy measured to forecast the class of test data based on the class information of training instances. It works with an assumption that the forecasted features are conditionally independent given the class. It makes sure that there is no influence of hidden attributes on the forecasting method [26].

E. Neural Networks

Neural networks have great importance for classification. Being self-adaptive procedures with data, they are able to fit themselves to the data without obvious description of underlying model's function [27]. They are adjustable in prototyping real world perplexing relationships because of being non-linear models. Moreover, they can estimate the posterior probabilities which helps to establish classification rule and perform statistical analysis [28].

F. Decision Tree

Decision tree creates a binary tree. When solving classification problem, decision tree procedure is very useful. A tree needs to be set up to prototype the classification procedure. After constructing the tree, it is needed to apply to each tuple of dataset. Then, it shows the result of classification for that tuple [29].

V. EXPERIMENTAL ANALYSIS

Cross validation is a famous process for evaluating the accomplishment of any model. K-fold cross validation

partitions data in k subsets. Dimensions of the subsets are (almost) equal. The classifier is trained k times in total. Every single time, one subgroup is taken as test data and other subgroups are utilized like training data. By such process every subgroup is employed as test data once [30].

We have used $k=10$, the value of the cost parameter $C=1.00$ and regression loss epsilon is $\epsilon = 0.10$ for SVM in this paper. As RBF is used, we have taken various values of kernel scale γ for measuring the performance of SVM for our dataset.

When using SVM, for $\gamma = 0.03$ we get the highest values for all. The classification accuracy for is 94.30% and the Area Under ROC Curve (AUC) is 0.979. The values of AUC are decreasing gradually as the value of γ increases that means the performance of SVM is better for lower kernel scale values for our dataset. The classification accuracy is also decreasing as kernel scale value increases. But for $\gamma = 0.88$ the classification accuracy increased suddenly from 68.50($\gamma = 0.88$) to 69.30 then again started to decrease. While comparing the machine learning techniques, we got the best classification accuracy for random forest classifier and that is 97.50%.

TABLE I. CLASSIFICATION ACCURACY, AREA UNDER ROC CURVE(AUC), F1, PRECISION AND RECALL FOR DIFFERENT KERNEL SCALE VALUES OF SVM

Serial No	Value of kernel scale γ	Classification Accuracy %	Area Under ROC Curve (AUC)	F1	Precision	Recall
1	0.03	94.30	0.979	0.942	0.943	0.943
2	0.13	87.00	0.905	0.870	0.870	0.870
3	0.29	86.30	0.883	0.862	0.862	0.863
4	0.43	72.30	0.881	0.700	0.736	0.723
5	0.58	70.00	0.874	0.669	0.716	0.700
6	0.70	68.50	0.861	0.641	0.714	0.685
7	0.88	69.30	0.855	0.651	0.722	0.693
8	0.95	66.70	0.844	0.616	0.691	0.667

TABLE II. THE COMPARISON OF CLASSIFICATION ACCURACY, AUC, F1, PRECISION AND RECALL FOR SVM, RANDOM FOREST, DECISION TREE AND NAÏVE BAYES CLASSIFIER

Classifiers	Classification Accuracy %	Area Under ROC Curve (AUC)	F1	Precision	Recall
SVM	94.30	0.979	0.942	0.943	0.943
Naïve Bayes	90.70	0.907	0.907	0.907	0.907
Random Forest	97.50	0.995	0.975	0.975	0.975
Decision Tree	95.80	0.973	0.958	0.958	0.958
Neural Network	93.50	0.983	0.935	0.935	0.935
kNN	85.50	0.915	0.854	0.855	0.855

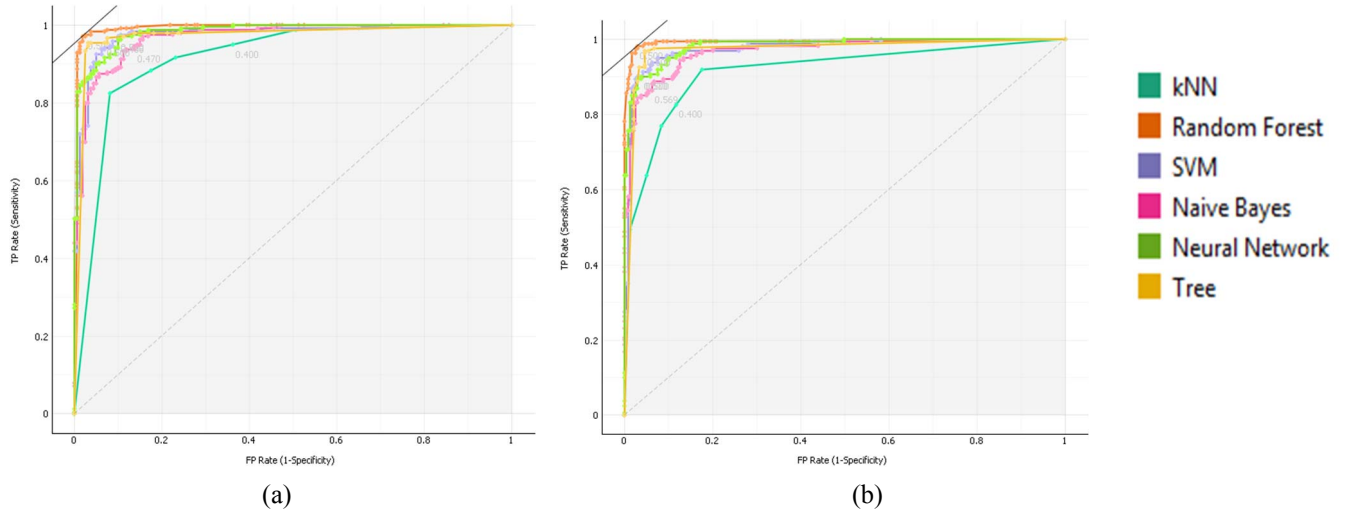


Fig.1. This figure shows the (a) ROC curve for 'yes' value detection and (b) ROC curve for 'no' value detection for kNN, Random Forest, SVM, Naïve Bayes, Neural Network and Decision tree.

VI. CONCLUSION

In this work, a different dataset creation method of organizing the online search log data has been used for liver cancer detection and also employed six machine learning techniques. We have evaluated the performance of SVM classifier for different kernel scale values for our dataset. We have also compared the classification accuracy of those machine learning techniques. A balanced dataset containing symptoms of liver cancer has been also produced for further researches. This method of organizing online search log data can be used in predicting other incurable diseases in future. An online prediction system can be built by applying the method. That online prediction system can easily predict the possibility of having a disease for a person and can give advance warning about presence of incurable diseases like cancers.

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